Unicode

In 1987 three engineers from Xerox (Joe Becker) and Apple (Lee Collins and Mark Davis) were tasked with answering a significant question: what if it was possible to produce a single encoding system that includes every character from every living language? (And later that would grow to include so-called "dead" languages, not to mention little pictures of smiling faces.) The answer was always "yes, provided you have enough bits"; the real challenge was finding a way to do it that didn't chew up vast amounts of disk and network unnecessarily. For example, if the sum total of characters across all languages is a big number like *250,000*, then we'd need a code unit minimum of 18 bits ($2^{18} = 262144$) in our code page to represent them all. Fine, but for English language text we already know we can get the job done with seven bits, so our hypothetical 18-bit encoding would have 11 zeroed bits on every Latin-1 character: around 63% of all Latin-1 files stored and transmitted in such a system would contain redundant zeroed bits. The bulk of text data for North American or Western European systems in such a system would therefore consist of redundant bits. Indeed, it was precisely this calculus that led to the original 7-bit design of ASCII63.

Growing beyond 8-bit codepoints was always going to be inevitable in such a system, but there already existed examples of 16-bit encoding solutions. Interoperability even between similar Western character sets was a growing challenge, but what about alphabets whose size exceeds a 256-point (8-bit) code page? Some of the oldest living languages in the world have alphabets measured in thousands of characters, after all. Various East Asian countries had rapidly growing computing needs in the latter half of the twentieth century, and with alphabets containing thousands of characters, they had little choice but to adopt a wider bit length long before it became a necessity in the Western Hemisphere. The GB 2312-1980 encoding standard covers the Simplified Chinese character set (it was originally a mere 6,763 Chinese characters, not including punctuation and other symbols), and the Japanese Industrial Standard (JIS) X 0208 specifies a double-byte encoding for Japanese characters (containing 6879 characters) and dates from 1978. There was also an international standard published in 1971 (ISO 2022) that set out ways in which computer systems could use double-byte or variable-width (variable numbers of bytes) in their character encoding.

Becker, Collins, and Davis conducted three main investigations: comparison between fixed and variable width encoding systems, impacts on storage requirements for 16-bit encoding, and estimates of character counts for all alphabets currently in use. From this the initial architecture of *Unicode* was derived. Unicode's name is